

Robotic Gripper Photo Labeling

<https://gemini.google.com/app/b6bbb84f5cb217ad>

User prompt: I'm labeling photos of physical objects for a computer-vision training dataset. The camera setup is stationary — it doesn't move between shots. A laser is projected onto each object as part of an object-identification aid, and lands on the same spot on the object across its repeat photos. Please ignore the laser dot/reflection itself when judging material appearance — it's not a feature of the object, just an identification aid from the capture rig. Look at the object in this photo and identify BOTH what the object is and what it's made of — even if this is difficult or you're not fully certain, give your best answer rather than skipping it. For material, give a GENERAL category rather than a precise guess at exact composition — e.g. "metal," "plastic," "rubber," "glass," "ceramic," "wood," "fabric," "composite" — not a specific alloy or polymer name. If the object is made of more than one material, list EACH material separately with which part of the object it belongs to, rather than picking just one. Only include a color reference when the color is a genuinely useful detail for telling this object apart from a similar one (e.g. distinguishing two otherwise-identical parts, or a color that's part of how the object is normally identified) — leave it out otherwise. Respond in two parts, in this exact order: 1. ONE plain-text line in this exact format: Object: <object> | Material: <material 1>, <material 2>, ... | Color: <only if relevant, else omit> 2. A JSON object in exactly this shape, with no other text before, between, or after the two parts: { "object": "<what the object is — be as specific as you reasonably can>", "materials": [{ "part": "<which part of the object this material applies to — use 'overall' if the whole object is one material>", "material": "<GENERAL material category, e.g. 'metal', 'plastic', 'rubber', 'glass', 'ceramic', 'wood', 'fabric', 'composite'>", "color": "<only if genuinely useful as a reference detail, else empty string>", "confidence": "high | medium | low" }], "labels": ["<label>", "<label>", ...], "notes": "<anything a human reviewer should double-check, or empty string>" } Guidelines: - Don't leave "object" or "material" vague just because you're unsure — give your best specific guess and reflect the uncertainty in "confidence" and "notes" instead of hedging in the field itself. - List every material you can distinguish, not just the most prominent one. - "labels" can be as long and detailed as useful — there is no length limit, include anything relevant (texture, finish, condition, shape, function, distinguishing features, etc.). - Set "confidence" to "low" per-material if that particular part is ambiguous, occluded, or hard to judge from the photo. - "notes" should flag anything a human reviewer should double-check (e.g. "could be plastic or coated metal," "handle partially out of frame"). This same object was photographed multiple times (laser landing on the same spot each time) — I'll apply your answer to all of those photos, so treat this as one label for the object as a whole, not just this one frame.

Response: Object: robotic gripper finger assembly | Material: plastic, metal | Color: blue { "object": "robotic gripper finger assembly", "materials": [{ "part": "fingers / flexible structural beams", "material": "plastic", "color": "blue", "confidence": "high" }, { "part": "linkage plates and fasteners", "material": "metal", "color": "", "confidence": "high" }], "labels": ["robotic gripper", "robot end effector", "soft robotic finger", "Fin Ray effect gripper", "blue plastic structural members", "metallic base plates", "screws and pivots", "articulated joint mechanism", "dual-finger gripper", "robotic hardware"], "notes": "The flexible blue finger structures appear to be 3D printed or molded plastic compliant mechanisms (Fin Ray structure), mounted to metal linkage plates with metal screws." }